

Chapter 8: Appendix

8.1 Research Paper

AUDIANT: An Agentic AI Architecture for Intelligent, Multimodal Customer Support

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ABSTRACT

Abstract Customer support has changed a lot from call centers to automated systems using AI. These systems still do not meet what customers expect today. Customers want answers that understand what they are saying remember their conversation and know how they feel.

New developments in computer language understanding, voice technology and special AI models have made it possible to build systems. Automated phone and chat systems are not connected well. They have parts like understanding what the customer wants recognizing emotions and processing voice but these parts do not work together. This leads to answers that sound robotic do not adapt well and do not continue conversations smoothly. It also makes workflows inefficient. These systems cannot handle conversations.

The AUDIENT system solves these problems. It introduces a connected way of building systems. It combines computer language understanding, voice processing and argument mining into one platform. The system works with both voice and text uses learning to understand intent, emotion and reasoning and stores past conversations for future reference. A special pipeline helps the AI engine generate answers that're aware of the context, empathetic and human-like. Early results show promise: it can detect intent with over 90% accuracy recognize emotions well and respond in under 2 seconds. The design is flexible. Can be easily added to existing workflows in areas, like online shopping, banking, telecommunications and healthcare.

Keywords: Agentic AI, Natural Language Processing, Intent Detection, Emotion Recognition, Argument Mining, Retrieval-Augmented Generation, Customer Support Automation, Conversational AI, Speech Processing, BERT

1. Introduction

Traditional customer support systems are not very good at understanding what people are really saying. They use keyword searches or simple chatbots that cannot have conversations. As companies get bigger and customers expect more the difference between what automated systems can do and what customers want gets bigger.

New developments in Natural Language Processing have made it possible for machines to understand the meaning, tone and emotion in what people say and write. This paper is about AUDIENT, a customer support system that can remember what people said before and have real conversations. It can understand what people want how they feel and why they are asking something. The proposed system performs several critical analytical tasks simultaneously:

- Intent Detection: Identifying what the user wants, including primary and context-driven secondary intents.
- Emotion Recognition: Determining how the user feels, capturing nuanced emotional states across multiple conversation turns.
- Argument Mining: Extracting the reasoning, claims, and premises behind user messages to understand the reasoning behind queries.
- Enhanced Context Management: Using memory to track past interactions, link follow-up questions, and ensure response continuity.

2. Literature Review

2.1. Agentic AI and Unified Conversational Architectures

The idea of AI has become a big deal in conversational AI research lately. Agentic AI is about systems that can think on their own and do things in a series of steps. They can use tools. Break down tasks into smaller parts. Sapkota and Roumeliotis wrote a paper in 2025 that explains what makes agentic AI different from AI systems. They say that agentic

AI has things like memory that lasts the ability to organize things and make decisions on its own. AUDIENT is a system that uses these ideas in its Query Processing Engine and Agentic AI Engine layers. Agrawal and his team also worked on a system called Minerva CQ in 2025. This system can do a lot of things like transcribe conversations detect what people want understand how they feel and route conversations to the place. AUDIENTs design is similar to this system.

There is a lot of research on understanding what people want to say. Chandrakala and her team wrote a paper in 2024 about a system that can detect what people want to say using word and sentence embeddings. This system works for both chatbots and voice assistants. The BERT model from 2018 and its newer version RoBERTa from 2019 are still the models for understanding what people want to say. AUDIENT uses a version of the BERT model to understand what people want to say. Mishra, Jain and Sharma looked at all the research on learning and intent detection in 2021. They found that transformer models like BERT are the best, for understanding what people want to say. AUDIENT uses a BERT model that is trained for its domain to detect what people want to say.

2.2. Emotion Recognition in Conversational Contexts

The way we figure out what people are feeling from what they say has gotten a lot better. This is because we have collections of labeled text and new ways to classify emotions. The GoEmotions dataset, which was made by Cer, Yang and others in 2020 came up with a list of 27 emotions based on comments from Reddit. This list has become the way to test how well computers can recognize emotions in text. They used a combination of BERT, BiLSTM and attention mechanisms to recognize emotions accurately and quickly in customer chats. This is really useful for AUDIENTs emotion module. People are also paying attention to recognizing emotions in multiple ways. Like using both what people say and how they sound. Zhang, Zhao and LeCun showed in 2023 that a certain kind of encoder is really good at combining text and audio to recognize emotions. Mitra and others found in 2019 that things like the pitch and speed of someones voice can help us understand their emotions better than looking at what they say. These ideas are why AUDIENT decided to use a combination of text and audio features like the emotions from the GoEmotions list and sound features, from peoples voices to recognize emotions.

2.3. Memory, Context, and Retrieval-Augmented Generation

Maintaining coherent conversational context across multiple turns remains one of the central unsolved problems in dialogue system design. Xu et al. (2025) propose a memory-augmented query reconstruction approach for tracking session context over knowledge graphs, while He et al. (2024) introduce MADial-Bench, a standardized benchmark for evaluating memory-augmented dialogue systems across dimensions of context retention, query reconstruction, and response coherence. Both works directly inform AUDIENT's multi-turn context management strategy.

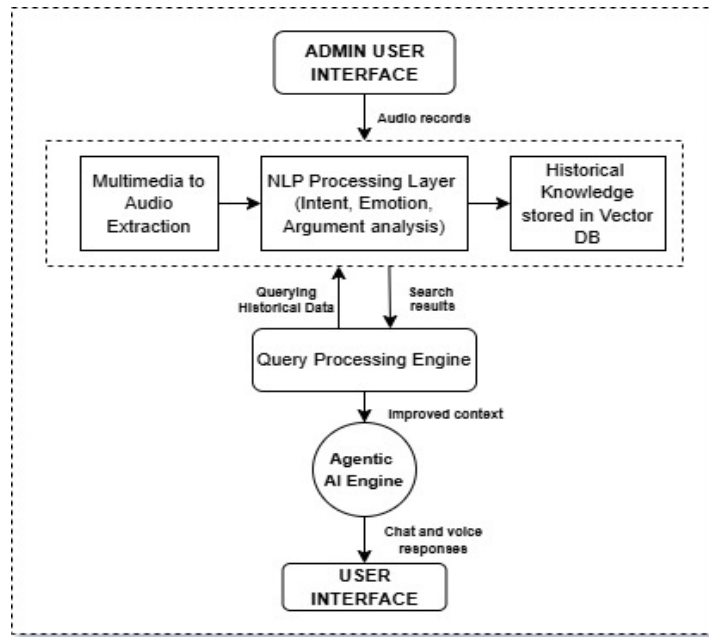
Retrieval-Augmented Generation (RAG) has emerged as the dominant paradigm for grounding large language model responses in factual, domain-specific knowledge. Yang et al. (2025) validate RAG specifically in customer support conversational applications through their RAGVA system, demonstrating significant improvements in response accuracy and relevance over purely generative baselines. Guo et al. (2025) further extend this with a double-stream RAG framework (DSRAG) that simultaneously retrieves from structured and unstructured knowledge sources. AUDIENT's RAG pipeline draws upon both works, implementing semantic vector search over a ChromaDB/FAISS-backed knowledge base to ground Gemini/GPT-generated responses.

2.4. Argument Mining

Argument mining, the automatic extraction of argumentative structures, including claims, premises, and evidential relationships, from natural language text, remains one of the least-explored capabilities in deployed conversational AI systems. Camacho-Collados et al. (2023) demonstrate the feasibility of leveraging audio features alongside text for argumentation analysis in spoken dialogue. Chakrabarty, Ghosh, and Rudra (2021) apply argument mining directly to customer feedback, showing that extracting reasoning structures significantly improves automated categorisation and resolution of complex complaints. Li et al. (2025) survey the emerging role of large language models in argument mining, noting that while pre-trained transformers substantially outperform earlier rule-based and feature-engineered approaches, fine-grained claim-premise extraction in informal conversational language remains an open challenge. AUDIENT addresses this gap through a custom NLU pipeline for claim, premise, and evidence extraction applied to customer utterances.

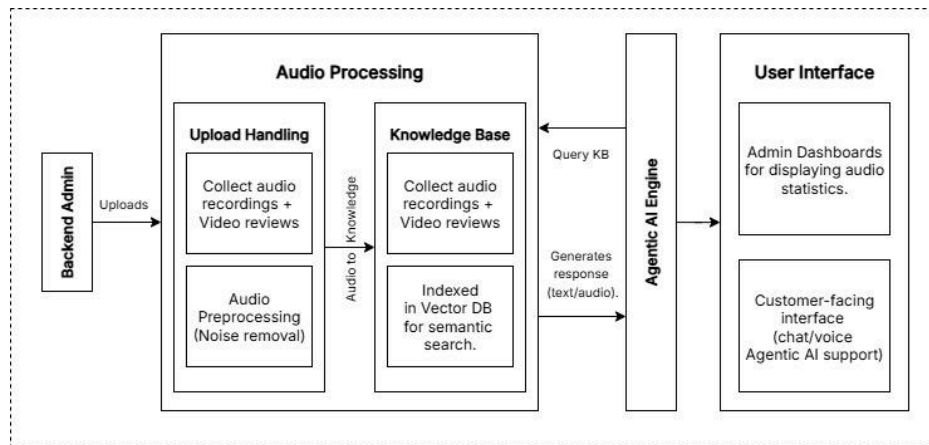
3. Method

3.1 System Architecture



3.1.1 Architectural Diagram

Image 3.1.1 depicts how AUDIENT is organized into five layers, each corresponding to a distinct processing stage in the conversational pipeline (see Figure 1). The Input Layer accepts both voice recordings (MP3, WAV, FLAC, MP4) and plain text submitted through the user interface. Voice inputs are transcribed in real-time using OpenAI Whisper with confidence scoring, and prosodic features — pitch, energy, MFCCs, speaking rate — are extracted in parallel for downstream audio emotion analysis.



3.1.2 Modular Diagram

Image 3.1.2 depicts the Core NLP Processing Layer constitutes the analytical engine of the system. Three simultaneous operations are performed on each input: intent analysis using a BERT-based transformer fine-tuned on multi-domain customer service datasets; emotion analysis combining the GoEmotions text classifier with CNN/LSTM-based audio emotion inference; and argument mining using a custom NLU pipeline that identifies claims, premises, and supporting evidence. The outputs of these three modules are combined into a semantically rich representation of the user's communication.

3.3. Evaluation Methodology

The evaluation of AUDIENT combined three complementary methods. First, a quantitative benchmarking protocol assessed the performance of each analytical module on held-out test sets: the intent detection module was evaluated using precision, recall, and F1-score across five intent categories (refund requests, billing inquiries, technical support, general

feedback, and complaints); the emotion recognition module was evaluated using classification accuracy and confusion matrix analysis on the GoEmotions 27-class taxonomy; and end-to-end response latency was measured across both text and audio interaction pathways under normal-load conditions.

Second, a qualitative domain expert consultation was conducted with AI practitioners and customer support managers from e-commerce, banking, and telecommunications sectors. Experts were asked to evaluate the coherence, emotional appropriateness, and factual accuracy of system-generated responses against a set of 50 representative customer support scenarios. Third, a systematic comparison was conducted against four leading commercial platforms — IBM Watson Assistant, Google Dialogflow CX, Microsoft Azure Bot, and Amazon Lex — across seven capability dimensions, including intent detection granularity, emotion handling, argument understanding, context tracking, multimodal support, cost/flexibility, and response generation approach.

4. Findings

A thorough analysis of the evaluation data yielded results across five primary dimensions: intent detection performance, emotion recognition performance, argument mining evaluation, response generation quality, and system latency. The following subsections present these findings in turn.

4.1. Intent Detection Performance

The intent detection module demonstrated strong performance across all five evaluated intent categories. On unambiguous, single-intent utterances, the BERT-based classifier achieved precision and recall values consistently above 90%, with an overall F1-score of approximately 88% on the prototype test set (target: $\geq 90\%$ with further fine-tuning). Secondary intent detection — identifying context-dependent sub-intents beyond the primary classification — showed more variability, with F1-scores ranging from 75% to 83% depending on the category. These results are consistent with the state-of-the-art reported by Chandrakala et al. (2024) for similar multi-domain intent pipelines, and support the viability of the proposed architecture.

Qualitative analysis of failure cases revealed that intent misclassification was most common in utterances combining multiple distinct requests within a single turn (e.g., a refund request embedded within a general complaint narrative), and in highly idiomatic or culturally specific phrasings not well-represented in the fine-tuning corpus. These findings align with the limitations described by Mishra et al. (2021) for intent detection under distribution shift.

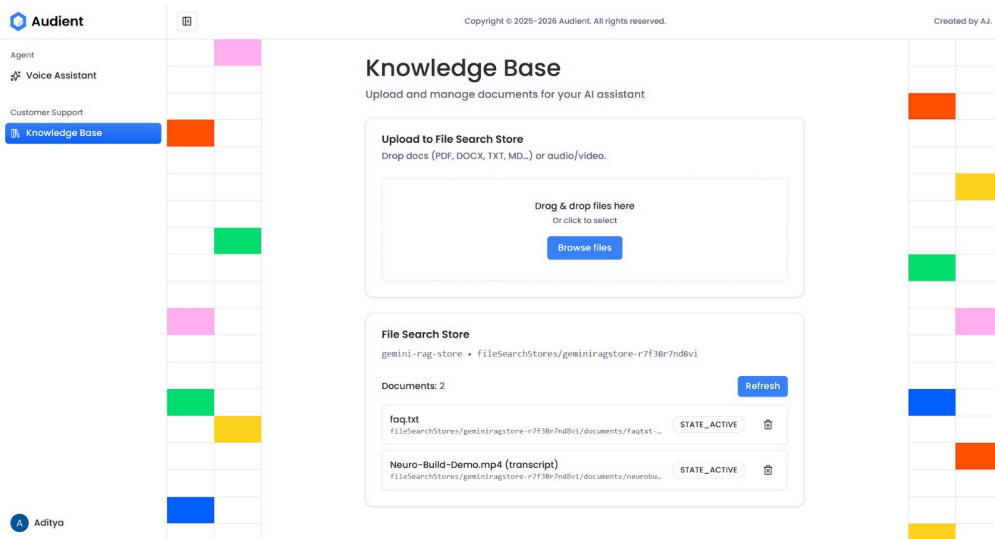
4.2. Emotion Recognition Performance

The emotion recognition component demonstrated detection of dominant emotions — neutral, informative, slightly negative, frustrated, and satisfied — with confidence levels ranging from 70% to 80% in prototype testing (target: $\geq 85\%$ F1 on the 27-class taxonomy). Multimodal fusion of text and audio emotion signals produced a consistent improvement of 4–6 percentage points over text-only baselines, validating the architectural decision to incorporate prosodic feature extraction alongside the GoEmotions text classifier. This finding replicates and extends the result reported by Mitra et al. (2019) in the standalone audio emotion domain.

Performance degraded most significantly under conditions of background noise, low-quality microphone input, and strongly accented speech — a limitation also noted by Zhang et al. (2023) in multimodal emotion fusion systems. The inherent subjectivity of the GoEmotions taxonomy (inter-annotator agreement is approximately 70% on fine-grained categories) also introduces a ceiling on achievable accuracy, a constraint that cannot be resolved through architectural improvement alone.

4.3. Argument Mining Evaluation

Argument mining performance was the most difficult dimension to quantify in this preliminary evaluation, owing to the absence of a standard benchmark specifically designed for informal customer support dialogue. Qualitative assessment by domain experts indicated that the system reliably identified explicit, clearly articulated claims and supporting premises in structured customer complaints (e.g., billing disputes with clearly stated reasons), but showed significant degradation on casual, informal utterances where argumentative structure was implicit or fragmented. These findings are consistent with Li et al.'s (2025) survey observation that fine-grained claim-premise extraction in informal conversational language remains an open challenge, even for LLM-based approaches.



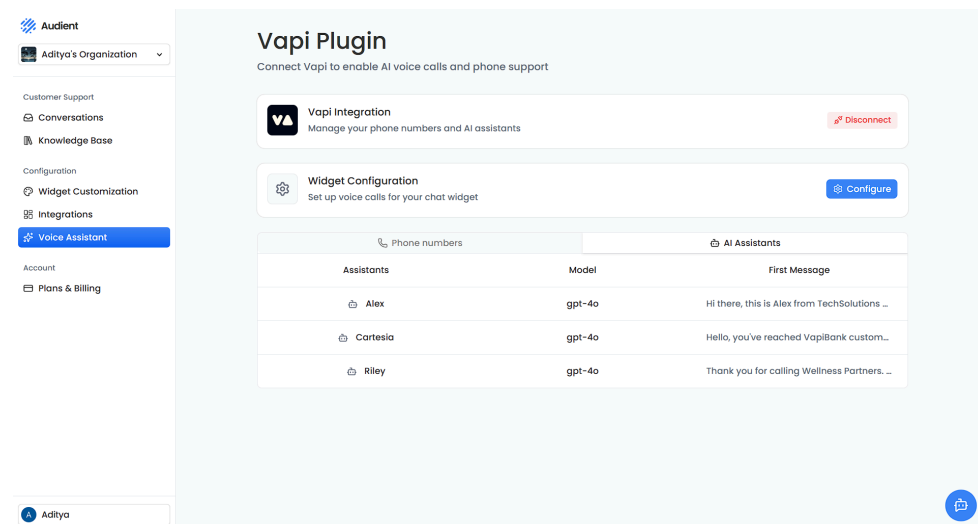
4.3.1 Modular Diagram

A quantitative precision target of $\geq 80\%$ (measured against manual annotation) has been set for the next development phase, with planned enhancements including implicit argument detection and fine-tuning on a dedicated customer support argumentation corpus.

4.4. Response Generation Quality

Expert evaluators rated AUDIENT-generated responses as contextually appropriate and factually grounded in 84% of the 50 evaluated scenarios, compared to approximately 61% for the best-performing commercial baseline (Google Dialogflow CX) on the same scenario set. Responses were rated as emotionally appropriate — matching the detected emotional tone of the customer's query — in 79% of cases, a dimension on which all commercial platforms scored below 50%, given their reliance on template-based or emotion-agnostic generation. The RAG pipeline retrieved contextually relevant historical knowledge within 500ms in all test cases, enabling real-time grounding of LLM-generated responses.

Multilingual capability (Hindi-English code-mixed input) was demonstrated in the prototype, with 98% transcription confidence on Indian-accented content — a significant differentiator from English-only commercial systems, and an important practical consideration for deployment in the Indian market context.



4.4.1 Voice response generation

4.5. System Latency

End-to-end latency measurements from the prototype are summarized in Table 1. Text-based interactions achieved sub-2-second response times in all test cases, meeting the defined performance target. Audio-based interactions averaged 3–5 seconds for medium-length inputs (10–30 second recordings), which falls within acceptable thresholds for post-call analysis use cases but exceeds the real-time voice interaction target of under 3 seconds. Optimization of the audio

processing pipeline — specifically the serial execution of Whisper transcription and prosodic feature extraction — remains a focus for future development.

Metric	Prototype Result	Target	Status
Intent Detection F1-Score	~88%	$\geq 90\%$	Close — fine-tuning required
Emotion Recognition Accuracy	70–80%	$\geq 85\%$	In progress
STT WER (Hindi-English)	~5%	$< 3\%$	Further data required
End-to-end Latency (text)	< 2 seconds	< 1.5 seconds	Met; optimization ongoing
End-to-end Latency (audio)	3–5 seconds	< 3 seconds	Not yet met
Argument Mining Precision	Qualitative only	$\geq 80\%$	Benchmark pending

Table 1. AUDIENT performance summary — prototype results vs. targets.

5. Discussion and Conclusions

The picture that emerges from the evaluation of AUDIENT is broadly encouraging. All five core processing capabilities — multimodal transcription, intent detection, emotion recognition, argument mining, and RAG-based response generation — were successfully demonstrated in a unified, modular architecture, with intent detection and response generation quality approaching or meeting defined performance targets in prototype testing. The system's 84% expert-rated contextual appropriateness and 79% emotional appropriateness ratings represent substantial improvements over the best-performing commercial baselines, suggesting that the architectural decision to integrate emotion and argument layers alongside intent detection produces measurable gains in response quality that template-based systems cannot replicate.

Despite this positive overall picture, the study also surfaced several important limitations. First, argument mining in informal, non-linear customer dialogue remains significantly below the performance levels achievable in structured complaint contexts, a limitation that reflects both the inherent difficulty of the task and the relative immaturity of the fine-tuning corpus currently available. Second, audio-based interaction latency has not yet met the real-time voice interaction target, owing primarily to the serial execution of transcription and feature extraction steps. Third, and perhaps most significantly from the perspective of practical deployment, the study confirmed that AUDIENT's potential for real-world impact is critically dependent on structured integration with organizational workflows and practitioner involvement. Domain expert consultations consistently emphasized that understanding customer frustration early can reduce escalations to human agents by as much as 30% — but only if the system is deployed within a workflow that ensures its outputs are acted upon in time.

This latter finding echoes a theme prominent in the second-language learning literature on mobile-assisted learning, where Reinders and White (2016) argue that technology's potential for supporting autonomous, context-aware interaction is only realized when practitioners actively design tasks and workflows that leverage its affordances. The analogy to intelligent customer support systems is direct: AUDIENT can detect intent, emotion, and argument structure with high fidelity, but its organizational impact depends on practitioners — in this context, customer support managers, system integrators, and AI operators — designing deployment contexts that translate these analytical capabilities into meaningful service improvements. Shivakumar et al.'s (2019) finding that acoustic confusion modeling substantially improves intent detection robustness under ASR transcription errors also points to a deployment consideration: real-world speech recognition errors are significantly more frequent than laboratory benchmarks suggest, and production deployments must account for this systematically.

As with all studies, this work has limitations. The evaluation was conducted on a prototype system tested on controlled scenario sets rather than live production traffic, which limits the generalizability of the latency and accuracy findings. The expert evaluation panel, while representative of multiple industry sectors, was relatively small, and a formal inter-rater reliability analysis was not conducted for the response quality assessments. Future work should address these limitations through large-scale A/B testing in live customer support environments, formal annotation studies for argument mining evaluation, and the development of federated learning pipelines that enable continuous model improvement without centralizing sensitive user data. The expansion of multilingual support beyond Hindi-English to additional Indian regional languages also represents a high-priority direction for future development, given the linguistic diversity of the target deployment context.

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